

What is Multivariate Data Analysis (MVDA)?



**Dr. Mahalanobis – Father of
Statistical Distance**

Univariate Data

- ❖ Dataset involving only one variable
- ❖ Example:
 - a) Students marks in class XII
 - b) Defects in machined products
 - c) Temperatures in the month of May in Chennai
 - d) Concentration of pollutant coming out of a factory near a river
- ❖ Data processed to find outliers, mean, median, mode, first quartile, second quartile

Bivariate Data

- ❖ Dataset involving two variables

- ❖ Example:

- a) Students marks in laboratory and coursework

- b) Temperature and pressure data collected in ammonia reactor in transient mode

- c) Temperatures and rainfall in the month of May in Chennai

- d) Employee experience and salary data

- ❖ Data also processed to find covariance, correlation coefficient

Bivariate Data

- ❖ Dataset involving two variables
- ❖ Example:
 - a) Students marks in laboratory and coursework
 - b) Temperature & pressure data collected in ammonia reactor in transient mode
 - c) Temperatures and rainfall in the month of May in Chennai
 - d) Employee experience and salary data
- ❖ Data also processed to find covariance, correlation coefficient, simple regression

Univariate and Bivariate Normal Distributions

<https://medium.com/data-for-tomorrow/univariate-normal-distribution-47519960625e>

<https://medium.com/@irenemarkelic/how-to-create-a-bivariate-normal-random-vector-from-two-independent-normals-f5ee1ef884ed>

<https://brilliant.org/wiki/multivariate-normal-distribution/>

Multivariate Data

- ❖ Dataset involving 2 or more variables
- ❖ Measured simultaneously during each observation
- ❖ For purpose of data analysis
- ❖ Multivariate data may also be multi-variety as data may be numerical (-1,2,100) or categorical (Yes, No; Male, Female, Gender Neutral;)

Multivariate Data

❖ Examples:

- a) **Heart disease** as a function of age, weight, cholesterol, pulse rate, sugar
- b) **BET surface area** as a function of pore size distribution, presence of micro-pores and mesopores, functional groups present, activation agent used
- c) **Insurance policy pricing**: age, sex, income, health status, average regional lifespan

Multivariate Data Analysis (MVDA): Business Implications

Businesses must

- a) Offer higher-quality products & services at lower costs
- b) React quicker to changing market conditions (supply and demand)
- c) Optimize workforce (quality as well as quantity)
- d) Maximize profits

To achieve the above

- ❖ knowledge creation and management are essential
- ❖ No dearth of information, but lack of insight may hinder timely intervention

Multivariate Data Analysis (MVDA): Business Implications

Businesses use the collected data intelligently to garner actionable insights for

- ❖ Improved decision-making
- ❖ Developing business strategy
- ❖ Predictive modeling, and
- ❖ Accumulate business intelligence,

Often used in sectors like marketing, finance, and healthcare

Role of MVDA in Data Mining

- ❖ Historic data is stored in data warehouses
- ❖ Some of that information may be analyzed and understood with simple statistics (mean, median, quartiles), but

Most of the data in real life requires involved **data mining** through **multivariate methods** to convert them into

- ❖ Data Mining: Extracting patterns in complex data using intelligent methods
- ❖ Actionable insight for extracting patterns
- ❖ So, the “pattern” extracted may be: cluster, anomaly, hidden relationship, or a predictive rule

Move from intuition/experience/sentiment to data driven strategies and decisions

Data Mining in a Chemical Plant

- ❖ In a chemical plant, sensors continuously measure pressure, temperature, flowrate, vibration, and humidity.
- ❖ Machine learning algorithms analyze these multivariate data and
- ❖ Identify abnormal patterns indicating equipment failure or reactor instability before an accident occurs.